Case Study 3

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The .Rmd **and** *.html (or .pdf) should be uploaded in TUWEL by the deadline. Refrain from using explanatory comments in the R code chunks but write them as text instead. Points will be deducted if the submitted file is not in a decent form.*

Declaration of AI usage

We made no use of any AI tools in order to complete the sub tasks of Case Study 3.

Data

Load the data set you exported in the final Task of Case Study 2. Eliminate all observations with missing values in the income status variable.

As a reminder, the data set includes world data from 2020, focusing on:

- **Education Expenditure (% of GDP)**
- **Youth Unemployment Rate (15-24 years)**
- **Net Migration Rate** (difference between the number of people entering and leaving a country per 1,000 persons)

for most world entities in 2020. The data was downloaded from https://www.cia.gov/the-world-factbook/about/archives/. Additional information on continent, subcontinent/region and income status was appended to the dataset in Case Study 2.

```
data_raw <- read_csv("file_out2.csv",
                     na = c(".", "NA", ""),
                     show_col_types = FALSE)

data <- data_raw %>%
  rename(income_status = status) %>%
  filter(!is.na(income_status)) %>%
  mutate(
    income_status = factor(income_status, levels = c("H", "UM", "LM", "L")),
    continent = factor(continent),
    subcontinent = factor(subcontinent)
  )

glimpse(data)
```

```
## Rows: 216
## Columns: 11
## $ ...1              <dbl> 1, 2, 3, 4, 5, 6, 8, 9, 10, 11, 12, 13, 14, 15, 16, ~
## $ country           <chr> "Afghanistan", "Albania", "Algeria", "American Samoa~
## $ ISO3              <chr> "AFG", "ALB", "DZA", "ASM", "AND", "AGO", "ATG", "AR~
## $ continent         <fct> Asia, Europe, Africa, Oceania, Europe, Africa, Ameri~
## $ subcontinent      <fct> Southern Asia, Southern Europe, Northern Africa, Pol~
## $ income_status     <fct> L, UM, LM, UM, H, LM, H, UM, UM, H, H, H, UM, H, H, ~
## $ expenditure       <dbl> 4.1, 3.6, NA, NA, 3.2, 3.4, NA, 5.5, 2.7, 5.5, 5.1, ~
## $ youth_unempl_rate <dbl> 17.6, 31.9, 39.3, NA, NA, 39.4, NA, 23.7, 36.3, NA, ~
## $ net_migr_rate     <dbl> -0.1, -3.3, -0.9, -26.1, 0.0, -0.2, 2.1, -0.1, -5.5,~
## $ low_yu            <chr> "no", "no", "no", NA, NA, "no", NA, "no", "no", NA, ~
## $ high_nmr          <chr> "no", "no", "no", "no", "no", "no", "yes", "no", "no~
```

```r
table(data$income_status)
```

```
##
##  H UM LM  L
## 79 54 56 27
```

Conclusion:

We used the provided file "file_out2.csv" provided in the TUWEL section of Assignment 3 and renamed status to income_status

Tasks:

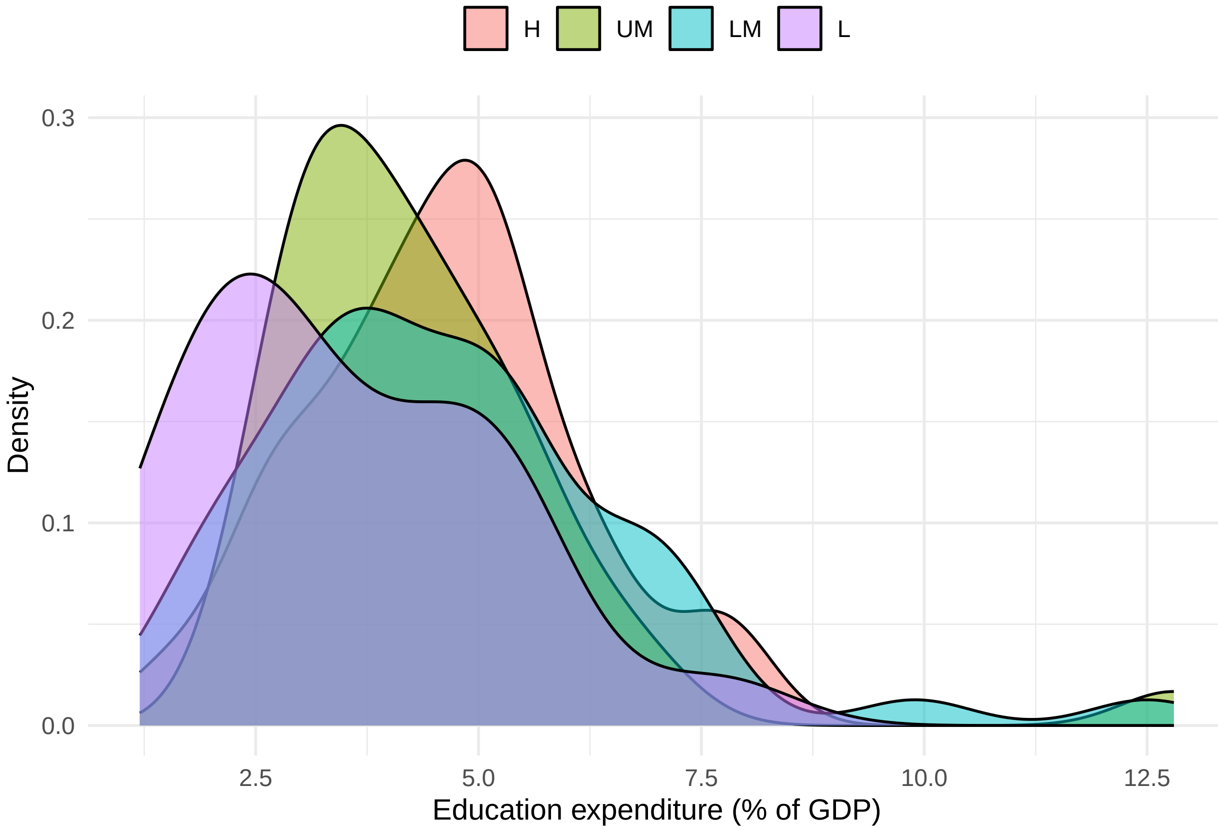
a. Education expenditure in different income levels

Using **ggplot2**, create a density plot of the education expenditure grouped by income status. The densities for the different groups are superimposed in the same plot rather than in different plots. Ensure that you order the levels of the income status such that in the plots the legend is ordered from High (H) to Low (L).

- The color of the density lines is black.
- The area under the density curve should be colored differently among the income status levels.
- For the colors, choose a transparency level of 0.5 for better visibility.
- Position the legend at the top center of the plot and give it no title (hint: use `element_blank()`).
- Rename the x axis as "Education expenditure (% of GDP)"

Comment briefly on the plot.

```r
ggplot(data, aes(x = expenditure, fill = income_status)) +
  geom_density(color = "black",
               alpha = 0.5,
               na.rm = TRUE) +
  labs(x = "Education expenditure (% of GDP)", y = "Density", ) +
  theme_minimal() +
  theme(legend.position = "top", legend.title = element_blank())
```



Conclusion:

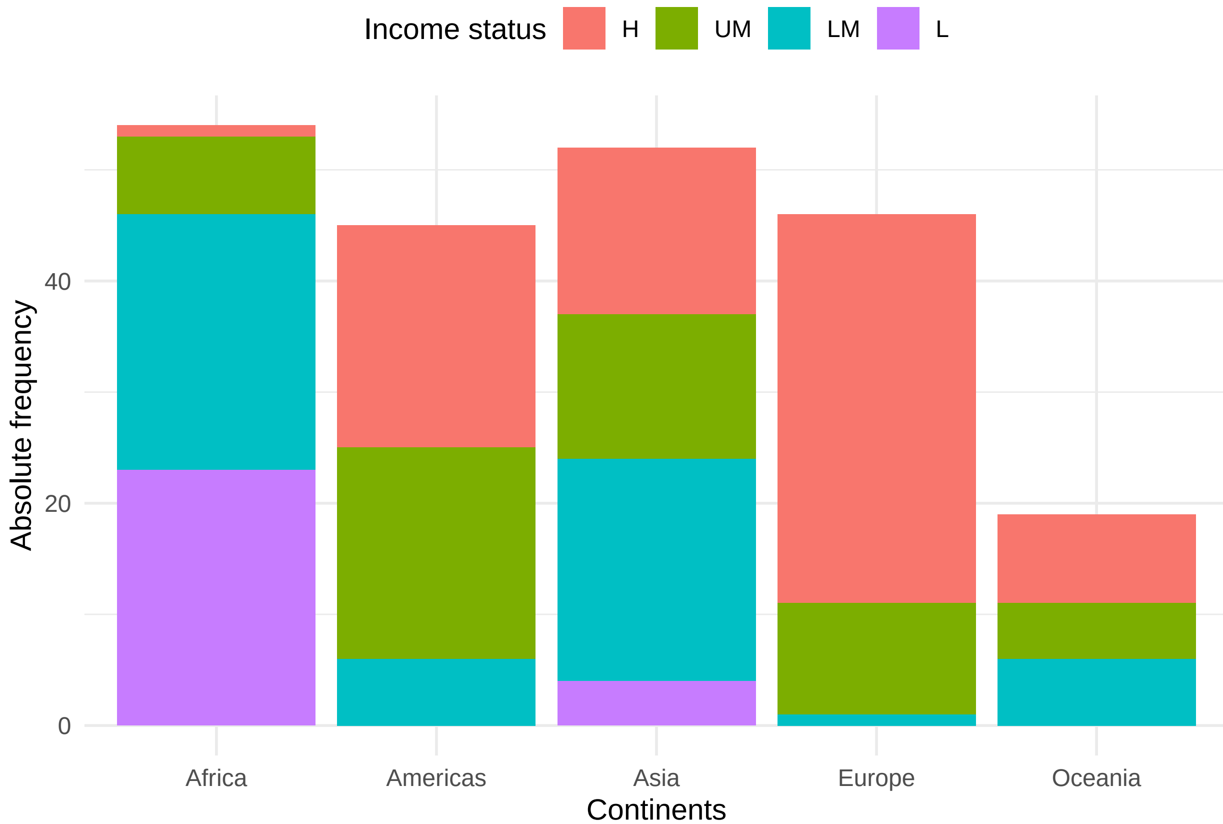
By inspecting the plot, we can see, that the education expenditure (% of GDP) strongly overlaps across all 4 income status groups e.g. across H, UM, LM and L. Most countries seem to spend roughly between 3% and 6% of GDP on education, so there seems not to be a clear separation between the income levels. Higher income countries seem to concentrate more around 5% of GDP, meanwhile lower income countries seem to gravitate more towards lower education expenditure (% of GDP). Overall, the differences between the income groups are not very pronounced.

b. Income status in different continents

Investigate how the income status is distributed in the different continents.

- Using **ggplot2**, create a stacked barplot of absolute frequencies showing how the entities are split into continents and income status. Comment the plot.

```
ggplot(data, aes(x = continent, fill = income_status)) + geom_bar() +
  labs(x = "Continents", y = "Absolute frequency", fill = "Income status") +
  theme_minimal() +
  theme(legend.position = "top")
```

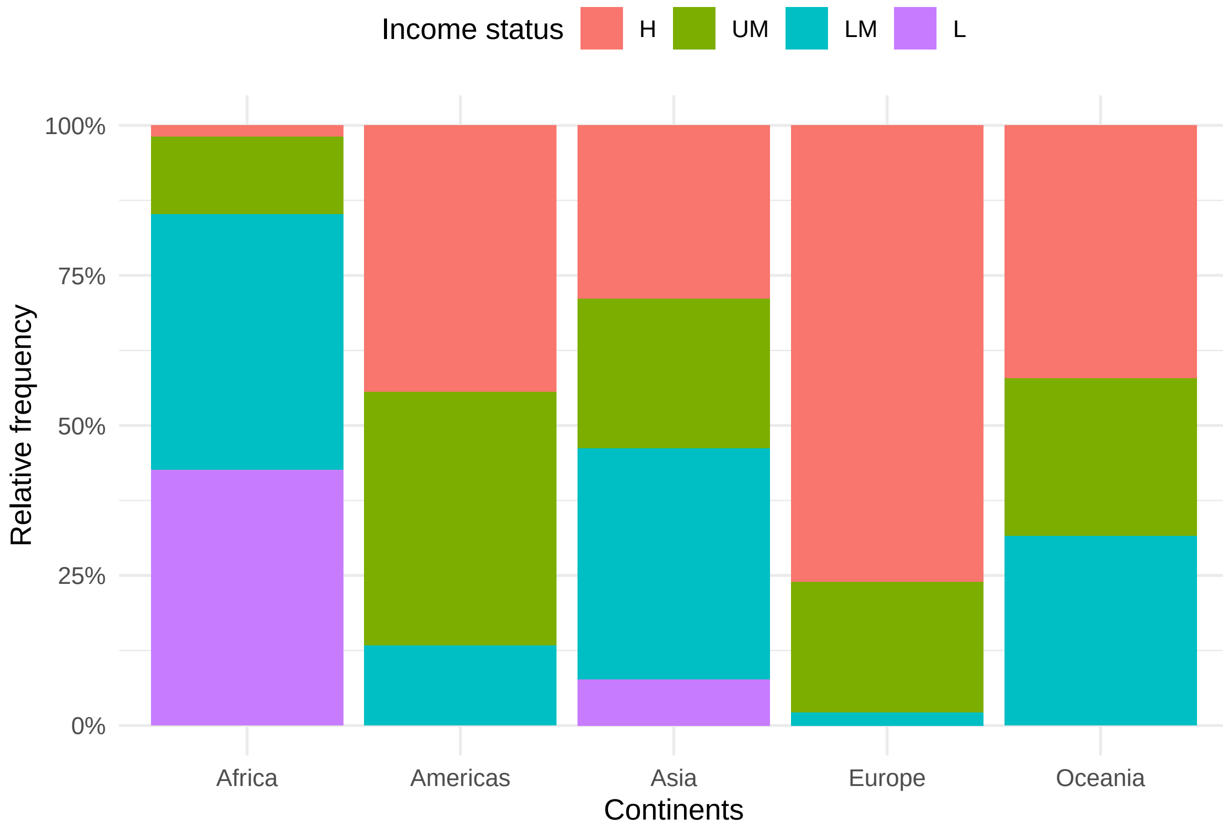


Conclusion:

The absolute frequency stacked barplots show that the number of entities differs strongly across continents. Africa and Asia contain many lower- and lower-middle income entities, while Europe seems to have the most high income entities. Asia and the Americas have a more mixed income distribution. Further Oceania seems to have fewer observations overall, as it's stacked barplot is compared to the others rather small.

- Create another stacked barplot of relative frequencies (height of the bars should be one). Comment the plot.

```
ggplot(data, aes(x = continent, fill = income_status)) + geom_bar(position = "fill") +
  scale_y_continuous(labels = scales::percent) +
  labs(x = "Continents", y = "Relative frequency", fill = "Income status") +
  theme_minimal() +
  theme(legend.position = "top")
```



Conclusion:

The stacked barplot of relative frequency makes the compositions of incomes across continents easier to compare, as Oceania is now easier to compare to the others. Europe has the largest proportion of high income entities, while Africa has the largest proportion of low- and lower-middle income entities. Asia and the Americas seem to be more mixed. Oceania mainly consists of lower-middle, upper-middle and high-income entities with no low-income entities.

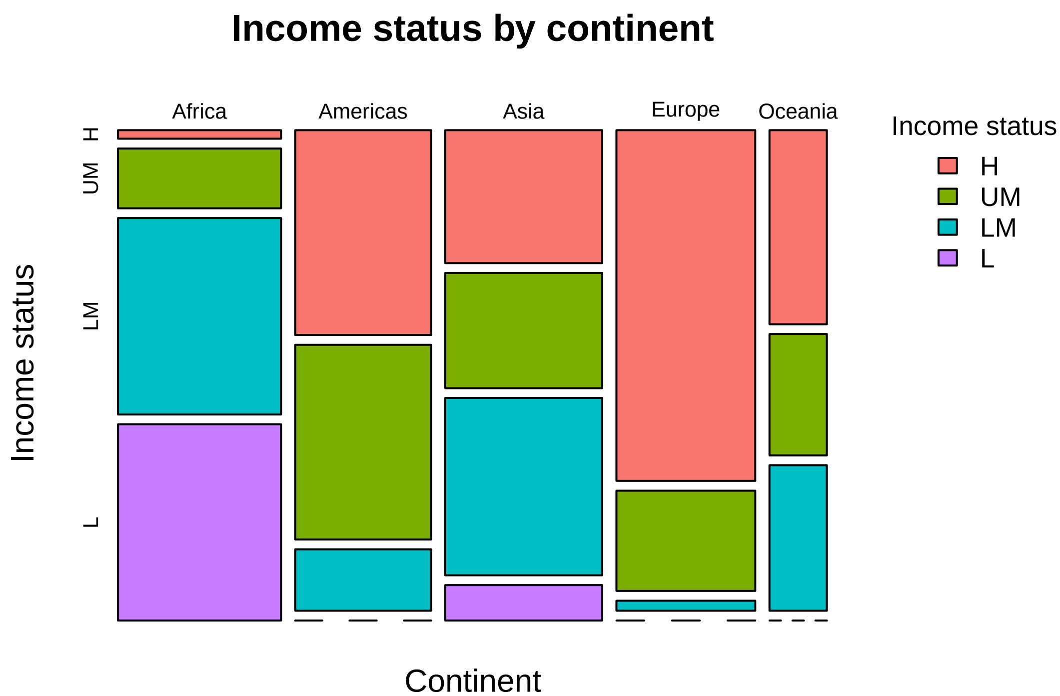
- Create a mosaic plot of continents and income status using base R functions.

```
income_continents <- table(data$continent, data$income_status)
income_status_colors <- c(
  "H" = "#f8766d",
  "UM" = "#7cae00",
  "LM" = "#00bfc4",
  "L" = "#c77cff"
)
par(mar = c(5, 4, 4, 8), xpd = TRUE)

mosaicplot(
  income_continents,
  color = income_status_colors,
  main = "Income status by continent",
  xlab = "Continent",
  ylab = "Income status"
)

legend(
```

```
  "topright",
  inset = c(-0.25, 0),
  legend = names(income_status_colors),
  fill = income_status_colors,
  title = "Income status",
  bty = "n",
  cex = 0.8
)
```



Conclusion:

By using a mosaic plot, we obtain a useful visual representation and overview of the income status of each continent and it is possible to recognize relationships between different variables. The width shows the number of observations per continent and the height of each colored part shows the income distribution within each continent.

- Briefly comment on the differences between the three plots generated to investigate the income distribution among the different continents.

Conclusion:

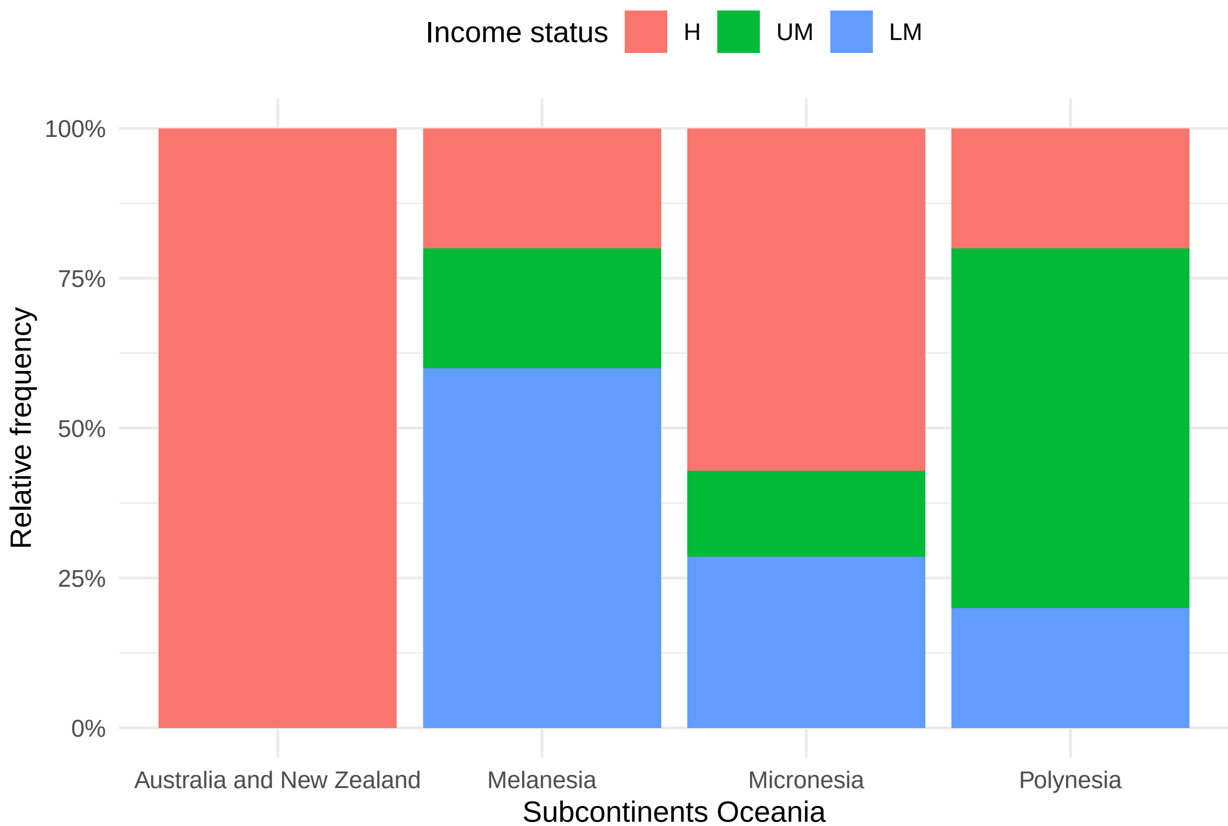
The absolute frequency stacked barplot is useful for comparing the total number in each continent, but in turn makes it harder to compare proportions as in the case of Oceania having a lower number of observations compared to the other continents. By using the relative frequency stacked barplot it is possible to remove the effect the number of observations of each continent have on size of each bar and therefore making income composition more clearly visible. Lastly the mosaic plot combines both features as mentioned before the width shows the number of observations, while the height of each colored part shows the income distribution within each continent.

c. Income status in different subcontinents

For Oceania, investigate further how the income status distribution is in the different subcontinents. Use one of the plots in b. for this purpose. Comment on the results.

```
oceania_data <- data %>%
  filter(continent == "Oceania") %>%
  mutate(subcontinent = fct_drop(subcontinent))

ggplot(oceania_data, aes(x = subcontinent, fill = income_status)) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = scales::percent) +
  labs(x = "Subcontinents Oceania", y = "Relative frequency", fill = "Income status") +
  theme_minimal() +
  theme(legend.position = "top")
```



Conclusion:

I used the relative frequency stacked barblot from subtask b. for further investigation, due to the subcontinents of Oceania having rather different numbers of observation. By doing that it is easier to compare proportions of income status between the subcontinents. We can see, that Australia and New Zealand only consist of high-income entities. Melanesia has the highest proportion of lower-middle-income entities, meanwhile Polynesia has the highest proportion of upper-middle income entities. Micronesia also seems to be more mixed but still with a high proportion of high-income entities. Also there are no low-income entities shown for Oceania in the data set.

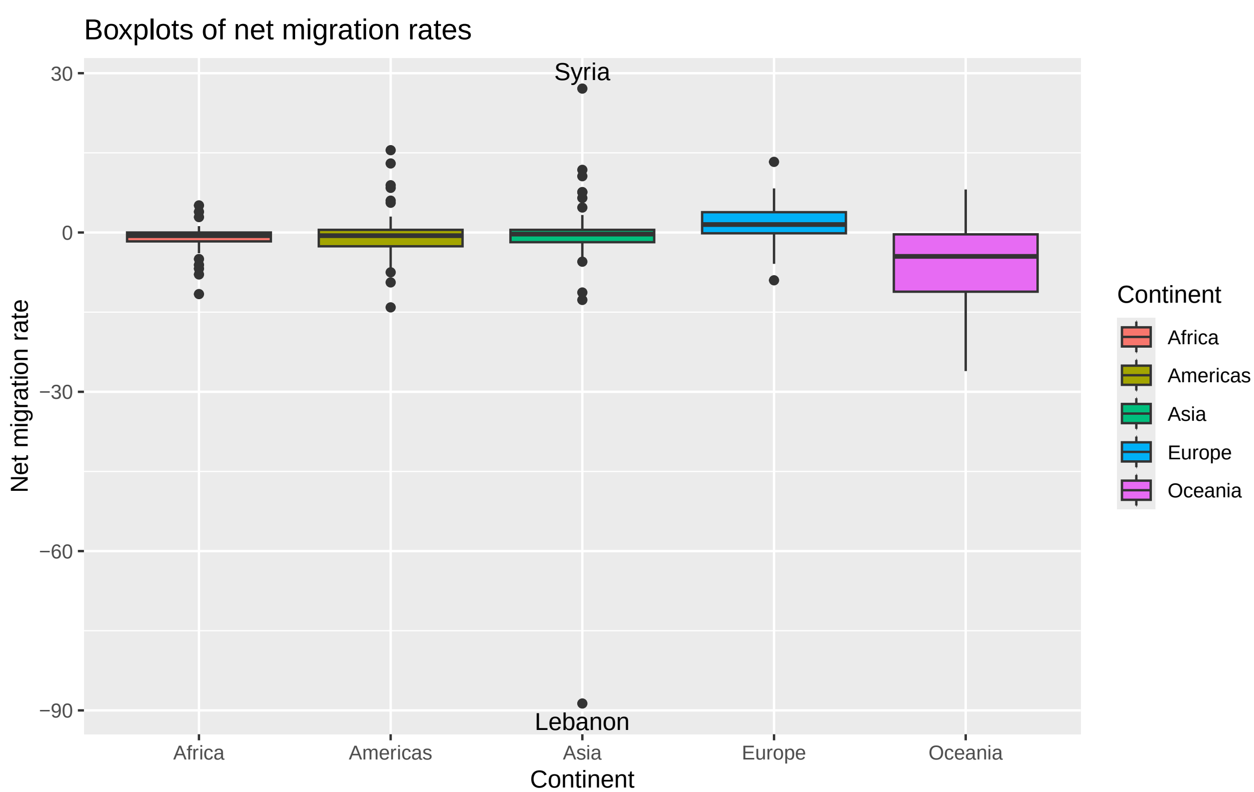
d. Net migration in different continents

- Using **ggplot2**, create parallel boxplots showing the distribution of the net migration rate in the different continents.
- Prettify the plot (change y-, x-axis labels, etc).

- Identify which country in Asia constitutes the largest negative outlier and which country in Asia constitutes the largest positive outlier.
- Comment on the plot.

```
outliers_df <- data %>% filter(continent == "Asia") %>%
  summarise(
    max_out = country[which.max(net_migr_rate)],
    max_val = net_migr_rate[which.max(net_migr_rate)],
    min_out = country[which.min(net_migr_rate)],
    min_val = net_migr_rate[which.min(net_migr_rate)]
  )

ggplot(data, aes(x = continent, y = net_migr_rate, fill = continent)) +
  geom_boxplot() +
  annotate(
    "text",
    x = "Asia",
    y = outliers_df$max_val,
    label = outliers_df$max_out, vjust = -0.5) +
  annotate(
    "text",
    x = "Asia",
    y = outliers_df$min_val,
    label = outliers_df$min_out, vjust = 1.5) +
  labs(title = "Boxplots of net migration rates",
       x = "Continent",
       y = "Net migration rate",
       fill = "Continent")
```



Conclusion:

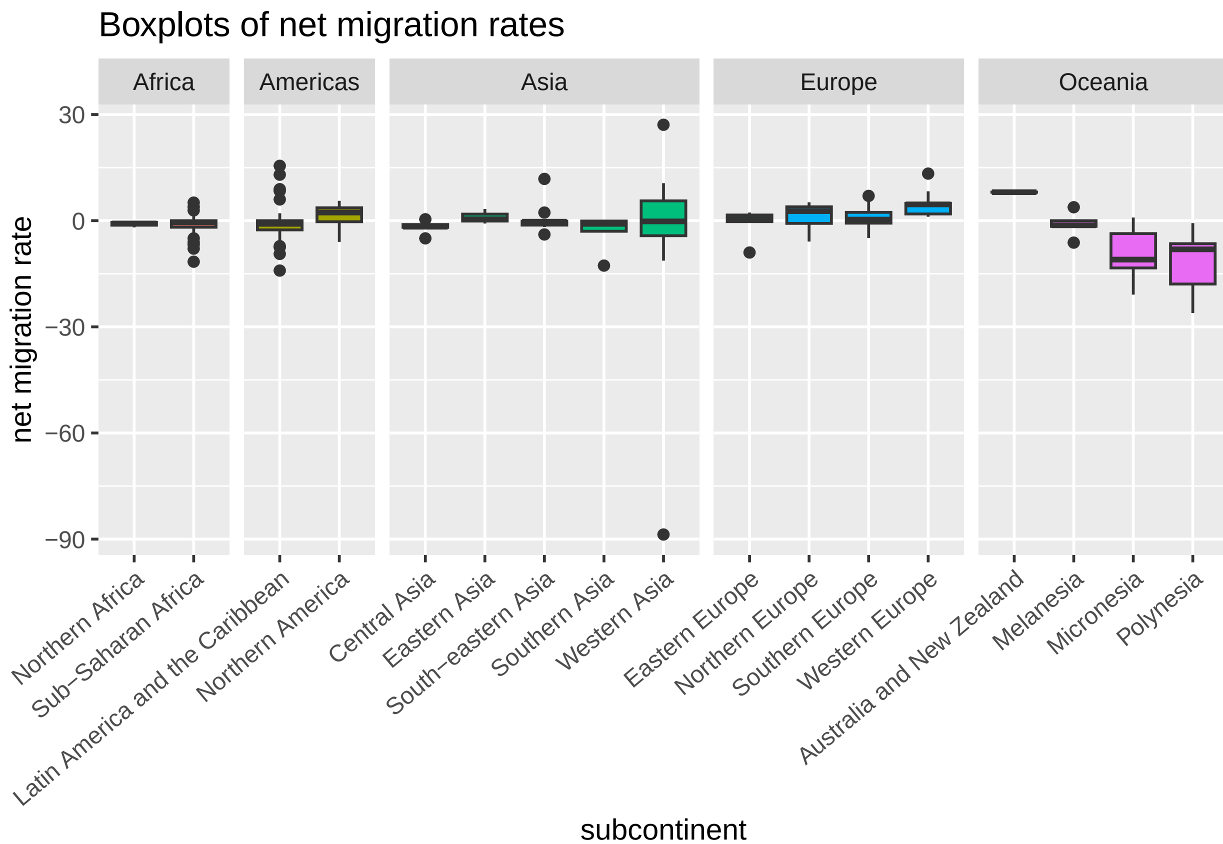
The largest positive outlier in Asia is Syria, the smallest Lebanon. Apart from Oceania all continents have distributions with median roughly at 0. Except for the extreme cases outliers roughly lie within +/- 15 of the median. By using annotate we are able to annotate the outliers directly.

e. Net migration in different subcontinents

The graph in d. clearly does not convey the whole picture. It would be interesting also to look at the subcontinents, as it is likely that a lot of migration flows happen within the continent.

- Investigate the net migration in different subcontinents using again parallel boxplots. Group the boxplots by continent (hint: use `facet_grid` with `scales = "free_x"`).
- Remember to prettify the plot (rotate axis labels if needed).
- Describe what you see.

```
ggplot(data, aes(x = subcontinent, y = net_migr_rate, fill = continent)) +
  geom_boxplot() +
  facet_grid(cols = vars(continent),
             scales = "free_x",
             space = "free_x") +
  labs(title = "Boxplots of net migration rates") +
  ylab("net migration rate") +
  theme(legend.position = "none",
        axis.text.x = element_text(
          vjust = 1,
          hjust = 1,
          angle = 40
        ))
```



Conclusion:

Under the assumption that migration mostly happens among geographical neighbors, we can now for example see a migration pattern from Latin America and the Caribbean to Northern America, although the continents total migration is roughly centered. Also Australia and New Zealand is the only subcontinent with a positive net migration in Oceania.

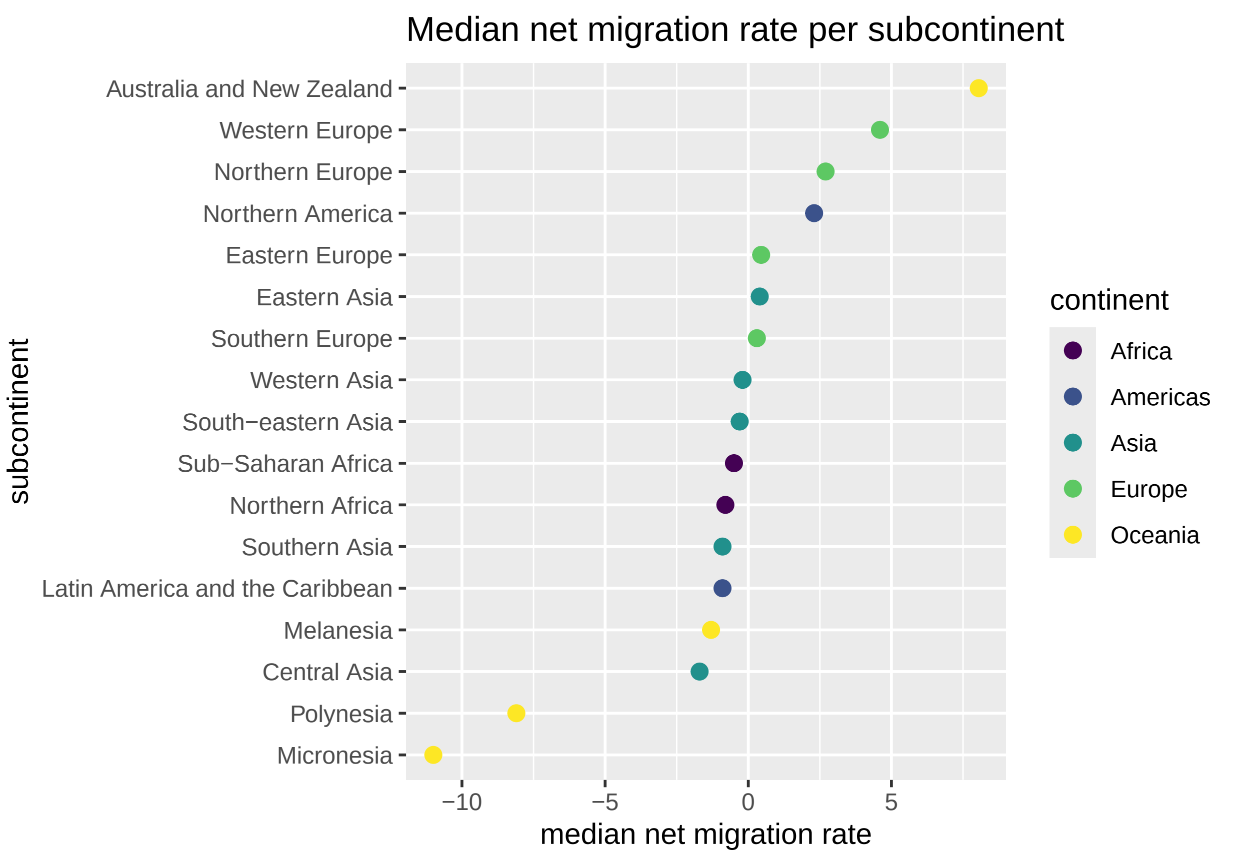
f. Median net migration rate per subcontinent.

The plot in task e. shows the distribution of the net migration rate for each subcontinent. Here you will work on visualizing only one summary statistic, namely the median.

For each subcontinent, calculate the median net migration rate. Then create a plot which contains the sub-regions on the y-axis and the median net migration rate on the x-axis.

- As geoms use points.
- Color the points by continent – use a colorblind friendly palette (see e.g., here).
- Rename the axes.
- Using `fct_reorder` from the **forcats** package, arrange the levels of subcontinent such that in the plot the lowest (bottom) subcontinent contains the lowest median net migration rate and the upper most region contains the highest median net migration rate.
- Comment on the plot. E.g., what are the regions with the most influx? What are the regions with the most outflux?

```r
data %>% group_by(subcontinent, continent) %>%
  summarise(median_net_migr_rate = median(net_migr_rate, na.rm = TRUE),
            .groups = "drop") %>%
  mutate(subcontinent = fct_reorder(subcontinent, median_net_migr_rate)) %>%
  ggplot(aes(x = median_net_migr_rate, y = subcontinent, color = continent)) +
  geom_point(size = 2.5) +
  labs(title = "Median net migration rate per subcontinent") +
  xlab("median net migration rate") +
  scale_color_viridis_d()
```



Conclusion:

Australia and New Zealand, Western Europe, Northern Europe and Northern America have the highest median influx, while Micronesia and Polynesia the highest median outflux. We also use the viridis palette for a color blind friendly color selection.

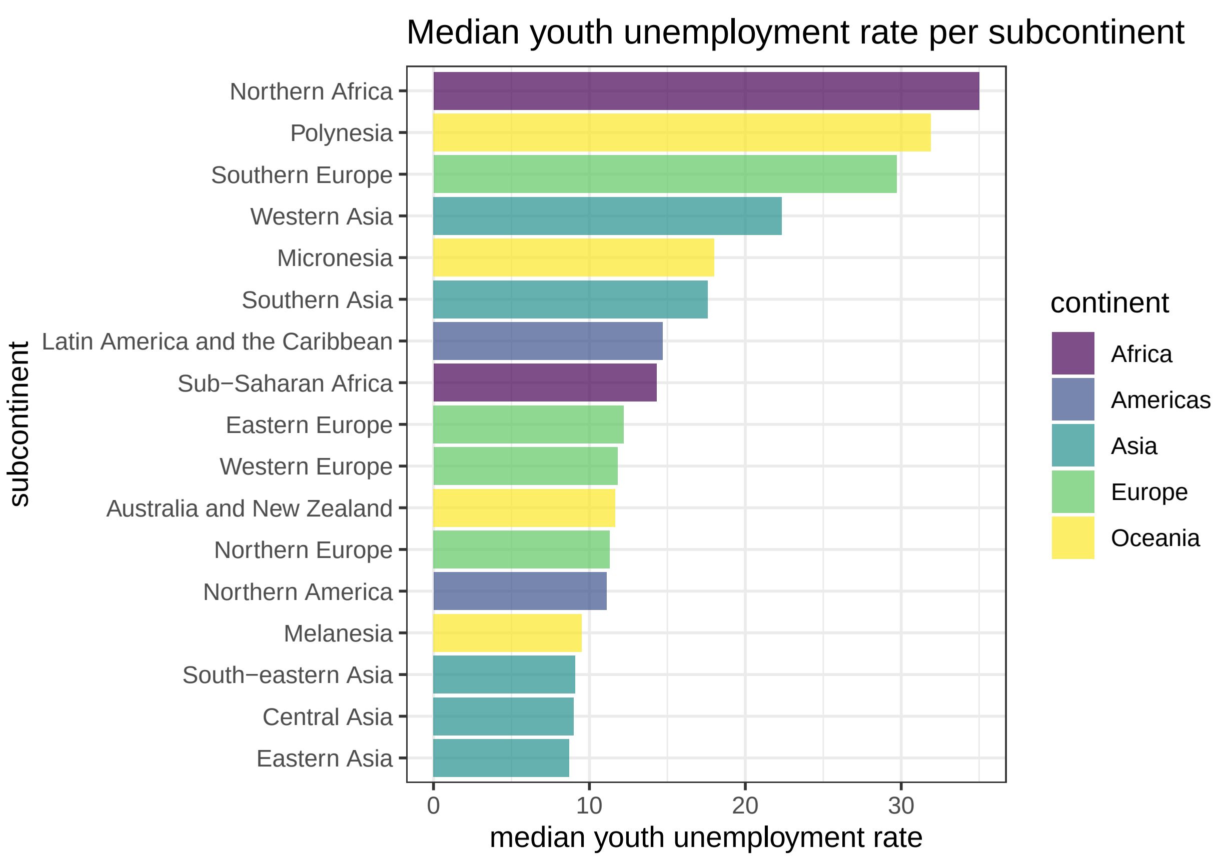
g. Median youth unemployment rate per subcontinent

For each subcontinent, calculate the median youth unemployment rate. Then create a plot which contains the sub-regions on the y-axis and the median unemployment rate on the x-axis.

- Use a black and white theme (`?theme_bw()`)
- As geoms use bars. (hint: pay attention to the statistical transformation taking place in `geom_bar()` – look into argument `stat="identity"`)
- Color the bars by continent – use a colorblind friendly palette.

- Make the bars transparent (use `alpha = 0.7`).
- Rename the axes.
- Using `fct_reorder` from the **forcats** package, arrange the levels of subcontinent such that in the plot the lowest (bottom) subcontinent contains the lowest median youth unemployment rate and the upper most region contains the highest median youth unemployment rate.
- Comment on the plot. E.g., what are the regions with the highest vs lowest youth unemployment rate?

```
data %>% group_by(subcontinent, continent) %>%
  summarise(
    median_youth_unempl_rate = median(youth_unempl_rate, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(subcontinent = fct_reorder(subcontinent, median_youth_unempl_rate)) %>%
  ggplot(aes(x = median_youth_unempl_rate, y = subcontinent, fill = continent)) +
  geom_bar(stat = "identity", alpha = 0.7) +
  labs(title = "Median youth unemployment rate per subcontinent") +
  xlab("median youth unemployment rate") +
  theme_bw() +
  scale_fill_viridis_d()
```



Conclusion:

The *stat = "identity"* argument in *geom_bar()* makes it behave like geom_col(), where instead of counting the number of categories in the categorical variable, it takes a second numeric variable, which it maps onto the bars length.

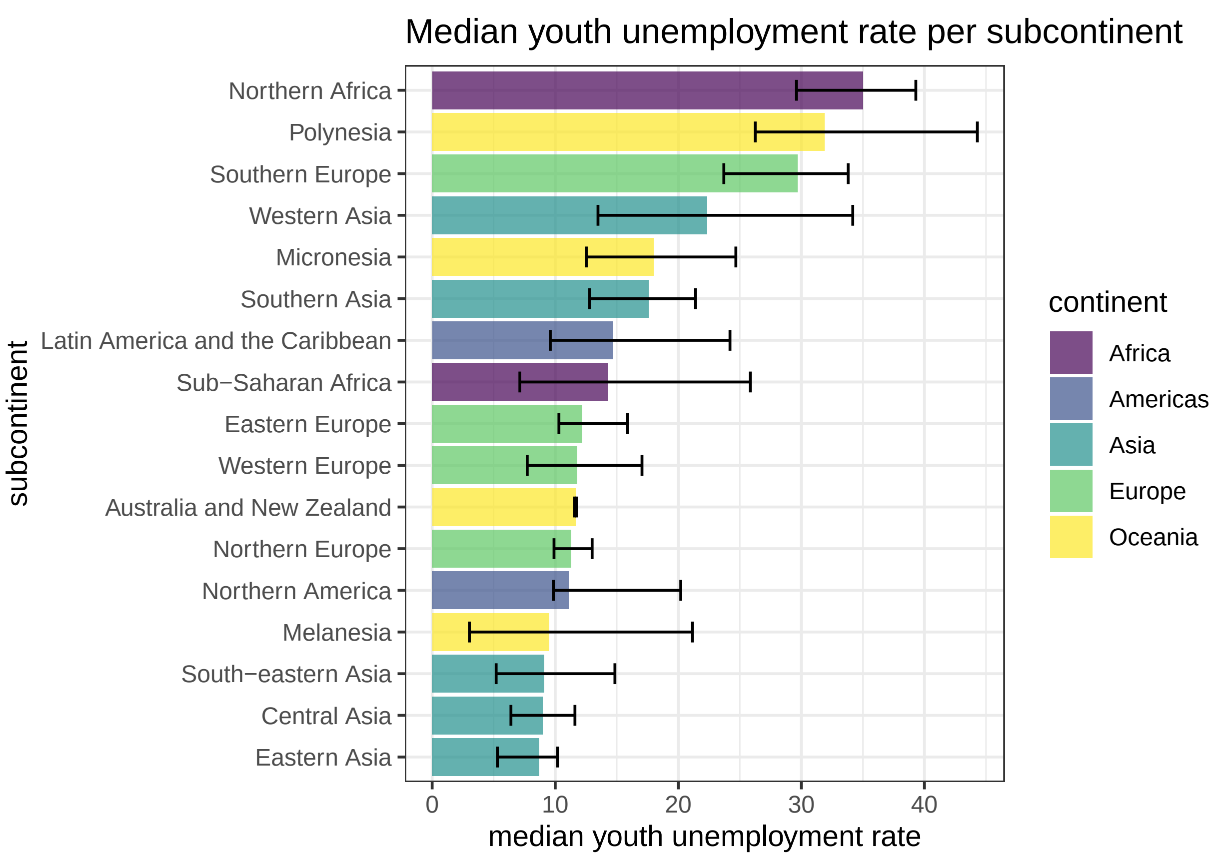
While Central, Eastern and South-eastern Asia have very low median youth unemployment rates, Northern Africa and Polynesia have the largest median youth unemployment rates.

h. Median youth unemployment rate per subcontinent – with error bars

The value displayed in the barplot in g. is the result of an aggregation, so it might be useful to also plot error bars, to have a general idea on how precise the median unemployment is. This can be achieved by plotting the error bars which reflect the standard deviation or the interquartile range of the variable in each of the subcontinents.

Repeat the plot from Task g. but include also error bars which reflect the 25% and 75% quantiles. You can use `geom_errorbar` in **ggplot2**.

```
data %>% group_by(subcontinent, continent) %>%
  summarise(median_youth_unempl_rate = median(youth_unempl_rate, na.rm = TRUE),
            Q1 = quantile(youth_unempl_rate, 0.25, na.rm = TRUE),
            Q3 = quantile(youth_unempl_rate, 0.75, na.rm = TRUE), .groups = "drop") %>%
  mutate(subcontinent = fct_reorder(subcontinent, median_youth_unempl_rate)) %>%
ggplot(aes(x = median_youth_unempl_rate, y = subcontinent, fill = continent)) +
  geom_bar(stat = "identity", alpha = 0.7) +
  geom_errorbar(aes(xmin= Q1, xmax = Q3), width = 0.5) +
  labs(title = "Median youth unemployment rate per subcontinent") +
  xlab("median youth unemployment rate") +
  theme_bw() +
  scale_fill_viridis_d()
```



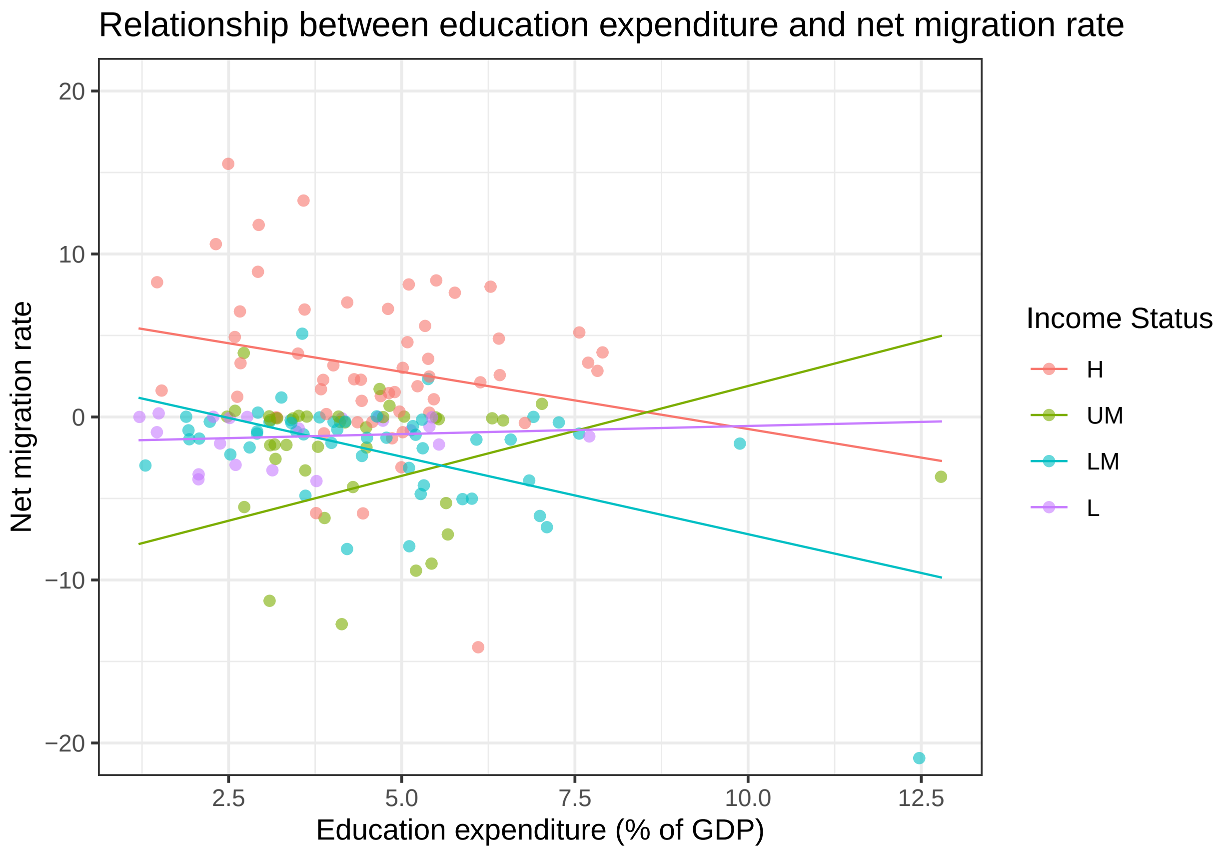
i. Relationship between education expenditure and net migration rate

Using **ggplot2**, create a plot showing the relationship between education expenditure and net migration rate.

- Color the geoms based on the income status.
- Add a regression line for each development status (using `geom_smooth()`).

Comment on the plot. Do you see any relationship between the two variables? Do you see any difference among the income levels?

```r
ggplot(data,
       aes(x = expenditure, y = net_migr_rate, color = income_status)) +
  geom_jitter(alpha = 0.6, na.rm = TRUE) +
  geom_smooth(
    method = "lm",
    formula = y ~ x,
    se = FALSE,
    fullrange = TRUE,
    alpha = 0.6,
    linewidth = 0.4,
    na.rm = TRUE) +
  coord_cartesian(ylim = c(-20, 20)) +
  labs(
    title = "Relationship between education expenditure and net migration rate",
    x = "Education expenditure (% of GDP)",
    y = "Net migration rate",
    color = "Income Status") +
  theme_bw()
```



Conclusion:

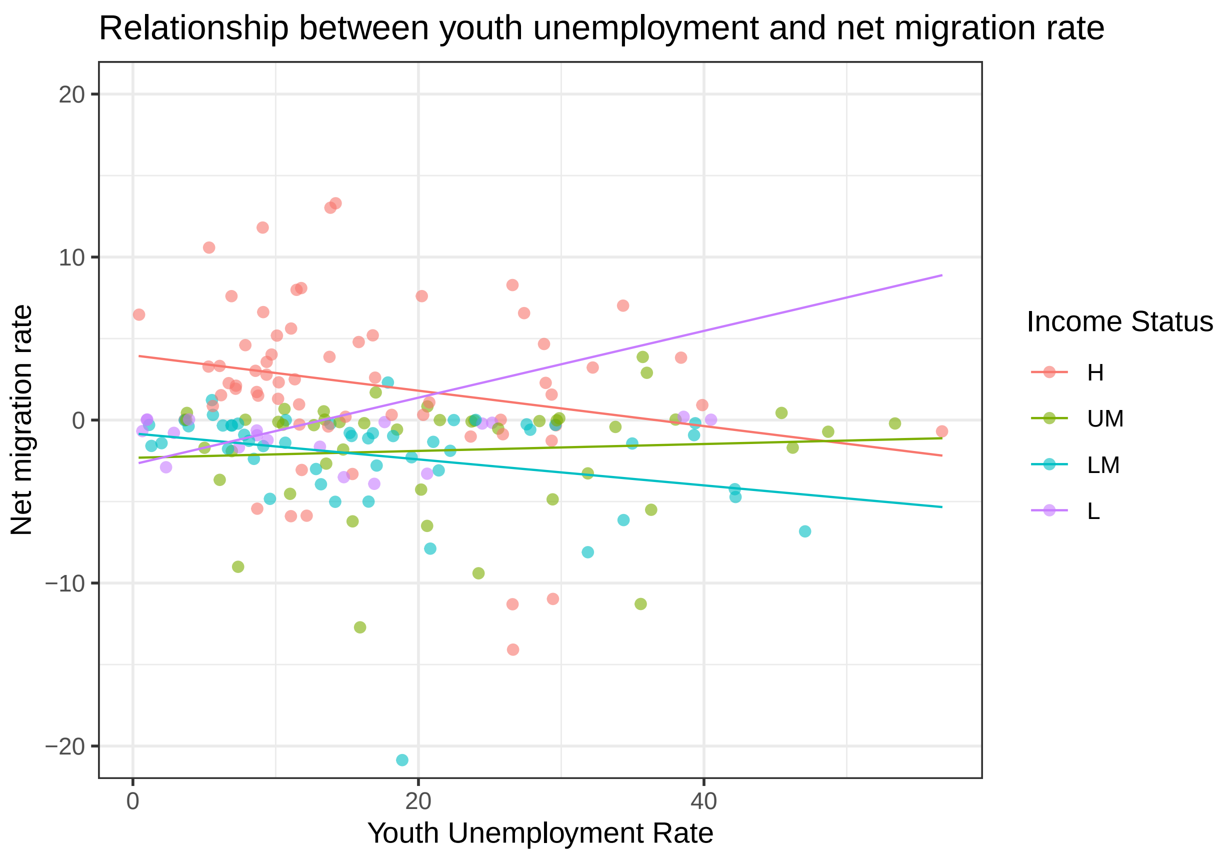
We set the y limits to -20 and 20 to make the plot more readable, (Lebanon and Syria are extreme outliers as previously seen in the boxplots). There seems to not be a strong relationship between education expenditure and net migration rate. The observations show a lot of variability across all levels of education expenditure. For high-income and lower-middle-income countries, the fitted lines indicate a slight downward trend, while upper-middle-income countries show a slightly positive trend. However, these differences should not be overinterpreted, since the data are highly scattered and the fitted lines are influenced by the outliers.

j. Relationship between youth unemployment and net migration rate

Create a plot as in Task i. but for youth unemployment and net migration rate. Comment briefly.

```
ggplot(data,
       aes(x = youth_unempl_rate, y = net_migr_rate, color = income_status)) +
  geom_jitter(alpha = 0.6, na.rm = TRUE) +
  geom_smooth(
    method = "lm",
    formula = y ~ x,
    se = FALSE,
    fullrange = TRUE,
    alpha = 0.6,
    linewidth = 0.4,
    na.rm = TRUE
  ) + coord_cartesian(ylim = c(-20, 20)) +
  labs(
    title = "Relationship between youth unemployment and net migration rate",
```

```
    x = "Youth Unemployment Rate",
    y = "Net migration rate",
    color = "Income Status"
  ) +
  theme_bw()
```



Conclusion:

We set the y limits to -20 and 20 to make the plot more readable, since Lebanon and Syria are extreme outliers as previously seen in the boxplots. Overall, there does not seem to be a strong relationship between youth unemployment and net migration rate. Most observations are concentrated at lower youth unemployment rates and around net migration rates close to zero. The fitted lines for high-, upper-middle- and lower-middle-income countries show only a slight downward trend. For low-income countries, the fitted line seems to increases, but there are still comparatively few observations in this group. Overall, the relationship appears weak and highly variable.

k. Merging population data

Go online and find a data set which contains the 2020 population for the countries of the world together with ISO codes.

- Download this data and merge it to the dataset you are working on in this case study using a left join. (A possible source: World Bank))
- Inspect the data and check whether the join worked well.

First we import the dataset and convert it to a tidy format:

```
population_url <- paste0(
  "https://api.worldbank.org/v2/country/all/indicator/SP.POP.TOTL",
  "?format=json&date=2020&per_page=20000"
)

population_json <- jsonlite::fromJSON(population_url)

population_data_20 <- as_tibble(population_json[[2]]) %>%
  transmute(
    ISO3 = countryiso3code,
    population2020 = value
  ) %>%
  filter(!is.na(ISO3), ISO3 != "", !is.na(population2020)) %>%
  distinct(ISO3, .keep_all = TRUE)

glimpse(population_data_20)
```

```
## Rows: 261
## Columns: 2
## $ ISO3           <chr> "AFE", "AFW", "ARB", "CSS", "CEB", "EAR", "EAS", "EAP",~
## $ population2020 <dbl> 694446100, 474569351, 453723239, 4459302, 101215763, 33~
```

Next we perform the left join with our current data:

```
data <- left_join(data, population_data_20, by = "ISO3")

glimpse(data)
```

```
## Rows: 216
## Columns: 12
## $ ...1              <dbl> 1, 2, 3, 4, 5, 6, 8, 9, 10, 11, 12, 13, 14, 15, 16, ~
## $ country           <chr> "Afghanistan", "Albania", "Algeria", "American Samoa~
## $ ISO3              <chr> "AFG", "ALB", "DZA", "ASM", "AND", "AGO", "ATG", "AR~
## $ continent         <fct> Asia, Europe, Africa, Oceania, Europe, Africa, Ameri~
## $ subcontinent      <fct> Southern Asia, Southern Europe, Northern Africa, Pol~
## $ income_status     <fct> L, UM, LM, UM, H, LM, H, UM, UM, H, H, H, UM, H, H, ~
## $ expenditure       <dbl> 4.1, 3.6, NA, NA, 3.2, 3.4, NA, 5.5, 2.7, 5.5, 5.1, ~
## $ youth_unempl_rate <dbl> 17.6, 31.9, 39.3, NA, NA, 39.4, NA, 23.7, 36.3, NA, ~
## $ net_migr_rate     <dbl> -0.1, -3.3, -0.9, -26.1, 0.0, -0.2, 2.1, -0.1, -5.5,~
## $ low_yu            <chr> "no", "no", "no", NA, NA, "no", NA, "no", "no", NA, ~
## $ high_nmr          <chr> "no", "no", "no", "no", "no", "no", "yes", "no", "no~
## $ population2020    <dbl> 39068979, 2528480, 44042091, 49761, 77380, 33451132,~
```

```
data %>%
  filter(is.na(population2020)) %>%
  select(country, ISO3, continent)
```

```
## # A tibble: 1 x 3
##   country ISO3  continent
##   <chr>   <chr> <fct>
## 1 Taiwan  TWN   Asia
```

Conclusion:

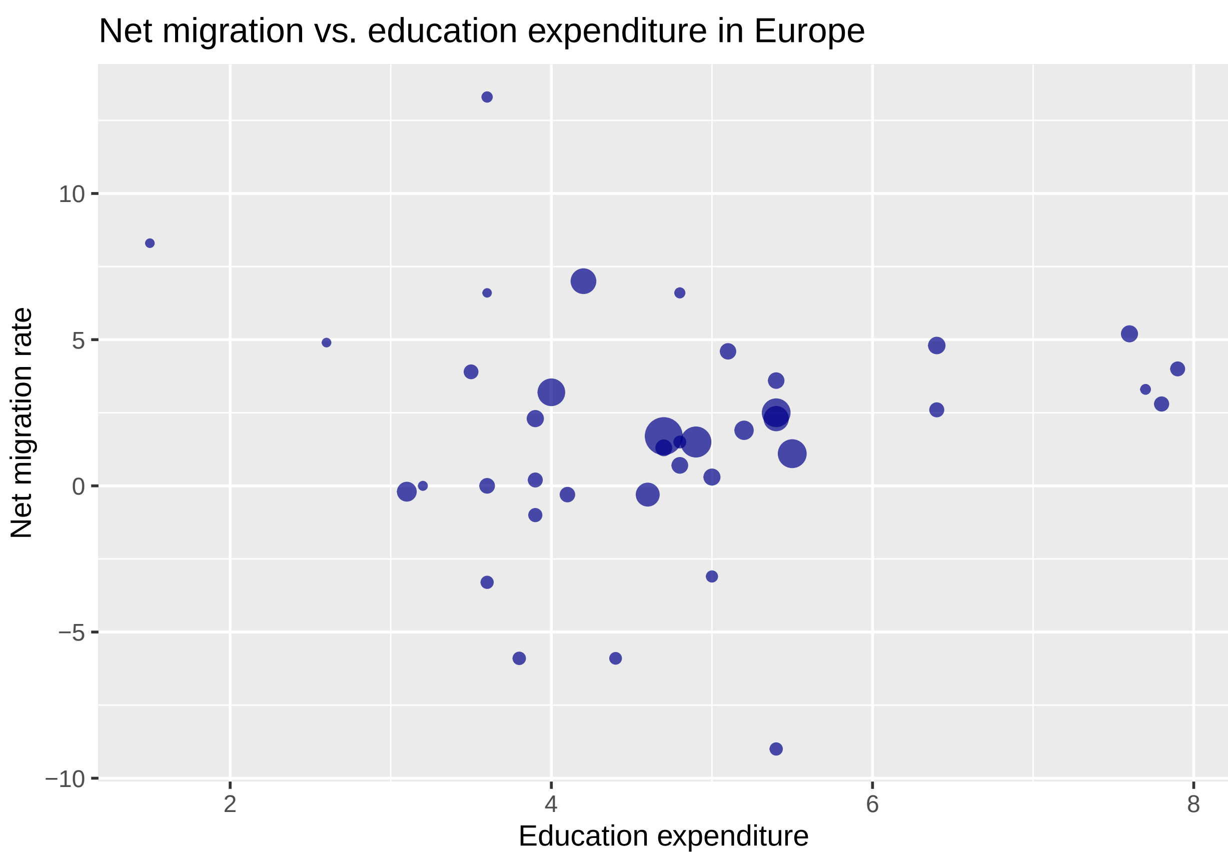
We have successfully added the population2020 column for all columns that were already present in our data except for the country Taiwan due to it not being matched and still having missing values in population 2020.

l. Scatterplot of education expenditure and net migration rate in Europe

Make a scatterplot of education expenditure and net migration rate for the countries of Europe.

- Scale the size of the points according to each country's population.
- For better visibility, use a transparency of `alpha=0.7`.
- Remove the legend.
- Comment on the plot.

```
data %>% filter(continent == "Europe" & !is.na(expenditure)) %>%
  ggplot(aes(x = expenditure, y = net_migr_rate,
             size = population2020)) +
  geom_point(alpha = 0.7,
             color = "darkblue") +
  labs(title = "Net migration vs. education expenditure in Europe",
       y = "Net migration rate",
       x = "Education expenditure") +
  guides(size = "none")
```



Conclusion:

One visible trend is that net migration rate increases with education expenditure. All the countries with the most education spending lie in the positive midfield of net migration rate. The most populated countries in Europe all seem to have a education expenditure between 4 and 6.

m. Interactive plot

On the merged data set from Task k., using function `ggplotly` from package **plotly**
re-create the scatterplot in Task l., but this time for all countries. Color the points according to their continent.

When hovering over the points the name of the country, the values for education expenditure, net migration rate, and population should be shown. (Hint: use the aesthetic `text = Country`. In `ggplotly` use the argument `tooltip = c("text", "x", "y", "size")`).

Hint: If you are creating a .pdf file, keep in mind that PDF output is static and does not support HTML widgets. To avoid build errors, you can wrap interactive plots in a conditional check or set the chunk option `eval=FALSE`. Alternatively, you can use the **webshot** package to automatically convert interactive elements into static images during the knit process.

```
plot_m <- ggplot(
  data,
  aes(
    y = net_migr_rate,
    x = expenditure,
    size = population2020,
    color = continent,
    text = country
  )
) + geom_point(alpha = 0.7) + labs(
  title = "Net migration vs. education expenditure in the World",
  x = "Education expenditure",
  y = "Net migration rate",
  color = "Continent"
) + guides(size = "none")
ggplotly(plot_m, tooltip = c("text", "x", "y", "size"))
```

Conclusion:

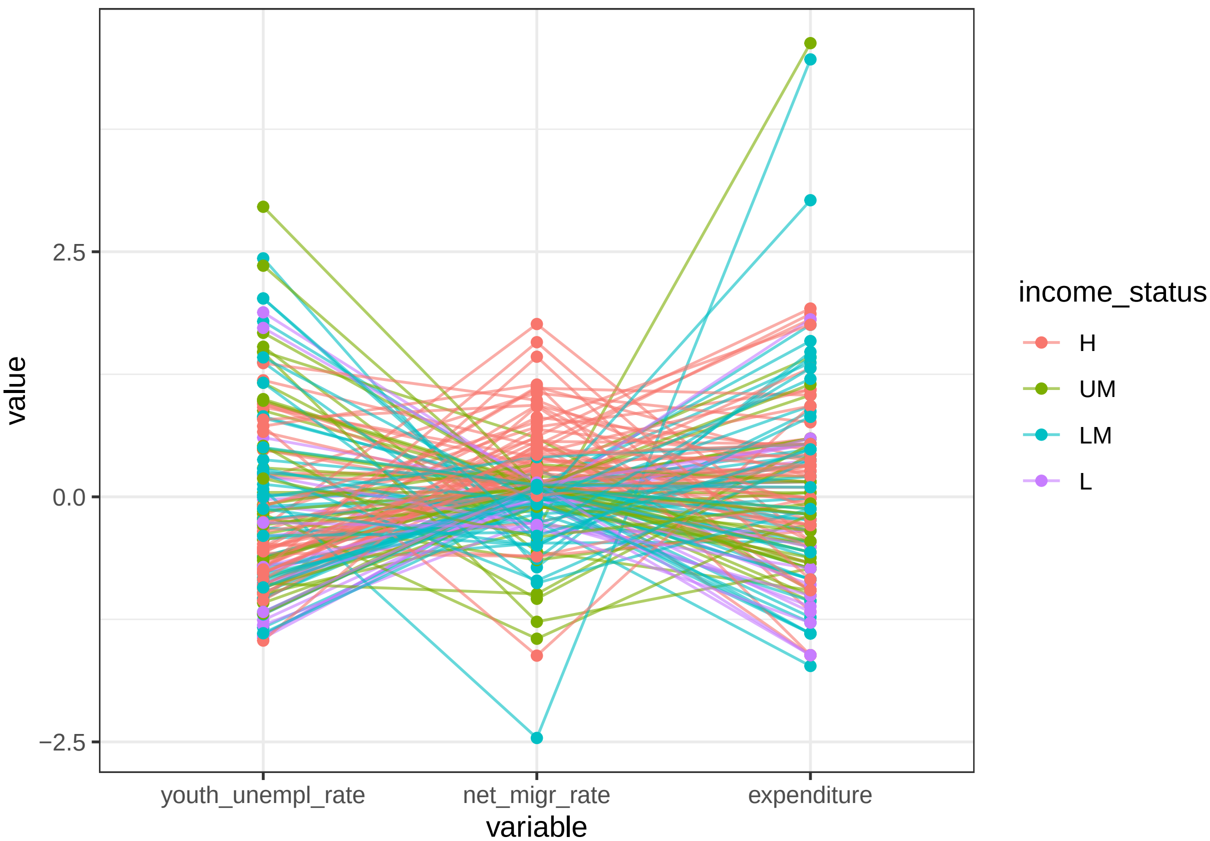
For analysis purposus we did not remove the outliers as in task i. We can now clearly and quickly see which countries produce interesting results.

n. Parallel coordinate plot

In **parallel coordinate plots** each observation or data point is depicted as a line traversing a series of parallel axes, corresponding to a specific variable or dimension. It is often used for identifying clusters in the data.

One can create such a plot using the **GGally** R package. You should create such a plot where you look at the three main variables in the data set: education expenditure, youth unemployment rate and net migration rate. Color the lines based on the income status. Briefly comment.

```r
ggparcoord(
  data,
  columns = c(8, 9, 7),
  groupColumn = 6,
  alphaLines = 0.6,
  showPoints = TRUE,
  scale   = "std"
)   +
  theme_bw()
```



Conclusion:

In a parallel coordinates plot, each observation is represented by a single line connecting all the measured values. We standardized the values for better comparison across collumns. Parallel lines indicate positive correlation between observations (e.g. countries in our case), while lines crossing in a "X" shape indicate negative correlation. Most noticably, there is a visble clustering of high income countries with high education expenditure, low youth unemployment and positive net migration. On the other hand, low income countries seem to cluster around low education expenditure, high youth unemployment and negative net migration. The middle income groups are more mixed but still show a tendency towards the same direction as the high and low income groups, respectivly.

o. World map visualisation

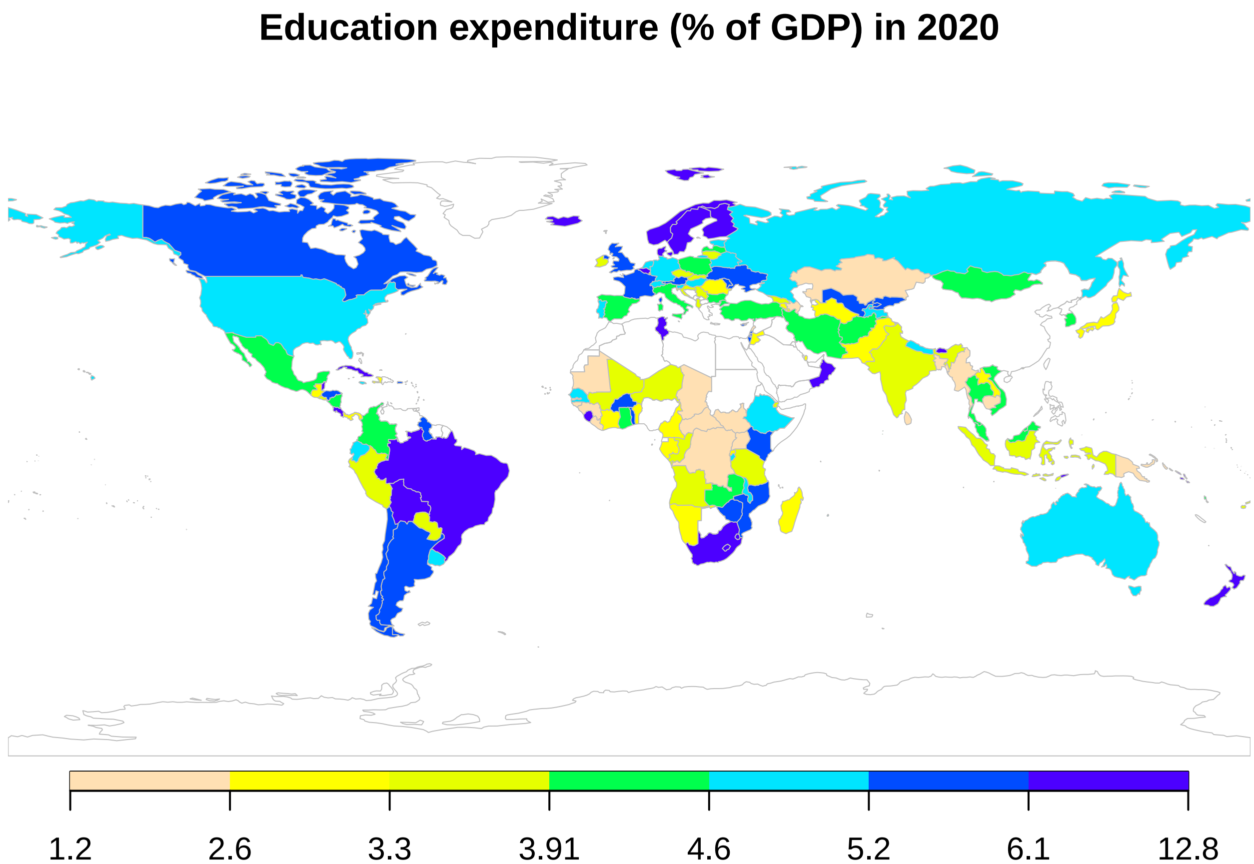
Create a world map of the education expenditure per country. You can use the vignette https://cran.r-project.org/web/packages/rworldmap/vignettes/rworldmap.pdf to find how to do this in R. Alternatively, you can use other packages (such as **ggplot2**, **sf** and **rnaturalearthdata**) to create a map.

```r
sPDF <- joinCountryData2Map(
  data,
  joinCode = "ISO3",
  nameJoinColumn = "ISO3",
  verbose = TRUE
)
```

```
## 215 codes from your data successfully matched countries in the map
## 1 codes from your data failed to match with a country code in the map
##      failedCodes
## [1,] "GIB"
## 29 codes from the map weren't represented in your data
```

```r
par(mai = c(0, 0, 0.2, 0),
    xaxs = "i",
    yaxs = "i")
mapParams <- mapCountryData(
  sPDF,
  nameColumnToPlot = "expenditure" ,
  mapTitle = "Education expenditure (% of GDP) in 2020",
  catMethod = "quantiles",
  colourPalette = "topo",
  addLegend = FALSE
)

do.call(
  addMapLegend,
  c(
    mapParams,
    legendWidth = 0.5,
    legendLabels = "all",
    legendIntervals = "page",
    legendMar = 2
  )
)
```



Conclusion:

We follow the guide in the vignete and plot a world map of education expenditure. In joining the data, only Gibraltar failed, which is no big problem!